

Name:

Maths Teacher:

SYDNEY TECHNICAL HIGH SCHOOL



Year 12

Mathematics Extension 1

HSC Course

Assessment 3

June, 2016

Time allowed: 90 minutes

General Instructions:

- Marks for each question are indicated on the question.
- Approved calculators may be used
- All necessary working should be shown
- Full marks may not be awarded for careless work or illegible writing
- ***Begin each question on a new page***
- Write using black or blue pen
- All answers are to be in the writing booklet provided
- A BOSTES Reference Sheet is provided with this Examination. Please do not write on it.

Section 1 Multiple Choice
Questions 1-5
5 Marks

Section II Questions 6-11
60 Marks

PART A: (5 Marks) Use the multiple choice answer sheet at the front of your Answer Booklet.

All questions are worth 1 mark

1	$\tan^{-1}(\sqrt{3}) =$ A. $\frac{\pi}{6}$ B. $\frac{\pi}{3}$ C. $-\frac{\pi}{3}$ D. $-\frac{\pi}{6}$
2	$\log_8 128 =$ A. e^{128} B. 16 C. $\frac{3}{7}$ D. $\frac{7}{3}$
3	$\sin(\cos^{-1}(-\frac{\sqrt{3}}{2})) =$ A. $-\frac{1}{2}$ B. $\frac{1}{2}$ C. $-\frac{\pi}{6}$ D. $\frac{\pi}{6}$
4	$\frac{d}{dx} (\log_2 3x) =$ A. $\frac{3}{x \ln 2}$ B. $\frac{3 \ln 2}{x}$ C. $\frac{1}{x \ln 2}$ D. $\frac{\ln 2}{x}$
5	$\tan(\cos^{-1} x) =$ A. $\frac{\sqrt{1-x^2}}{x}$ B. $\frac{-\sqrt{1-x^2}}{x}$ C. $\frac{x}{\sqrt{1-x^2}}$ D. $\frac{-x}{\sqrt{1-x^2}}$

PART B

(START EACH QUESTION ON A NEW PAGE)

QUESTION 6: (10 Marks)

Marks

(a) Find indefinite integrals of:

3

(i) $\frac{1}{x^2+9}$

(ii) $\frac{1}{1-3x}$

(iii) $\tan^2 x$

Find: (i) $\frac{d}{dx} \left(\frac{\ln x}{x} \right)$ (ii) $\frac{d}{dx} e^{\sin x}$

2

(c) Find the exact value of $\int_0^2 \frac{2 \, dx}{\sqrt{4-x^2}}$

2

(d) Find the second derivative of e^{x^3}

3

QUESTION 7: (10 Marks) Start a New Page

Marks

- (a) (i) Find the largest Domain, containing the point (4, 4) for which $f(x) = (x - 2)^2$ has an inverse function. **1**
- (ii) Sketch $y = f^{-1}(x)$ where $f^{-1}(x)$ is the inverse function defined in part (i) **2**
- (iii) State the Domain and Range of $f^{-1}(x)$ **2**

- (b) Sketch the graph of $y = 2\cos^{-1}\frac{x}{2}$ **2**

- (c) (i) Using one set of axes, neatly sketch the graphs of $y = e^x$ and $y = e^{-x}$. **3**
- (ii) On the same set of axes, use part (i) to sketch $y = \frac{1}{2}(e^x + e^{-x})$
(clearly label this graph)

QUESTION 8: (10 Marks) Start a New Page

Marks

(a) Solve for x : $\ln(x + 1) = 5$, giving your answer correct to 3 dec. places **1**

(b) The area under $y = \frac{1}{\sqrt{4+x^2}}$ and between the lines $x = 0$ and $x = 2\sqrt{3}$ is rotated about the x -axis. **3**

Find the exact volume of the solid formed.



(c) (i) Find $\frac{dy}{dx}$ if $y = x \tan^{-1} x$ **1**

(ii) Hence show that $\int_0^1 \tan^{-1} x \, dx = \frac{\pi}{4} - \frac{1}{2} \ln 2$ **3**

(d) (i) Find $\frac{d}{dx} \{ \sin^{-1} x + \cos^{-1} x \}$ **1**

(ii) What does this imply about the value of the expression $\sin^{-1} x + \cos^{-1} x$ as x varies over the Domain $-1 \leq x \leq 1$? **1**



QUESTION 9: (10 Marks) Start a New Page

Marks

(a) Show that $\frac{d}{dx} \ln\left(\frac{\sqrt{x-1}}{x}\right) = \frac{2-x}{2x(x-1)}$

2

- (b) The radius of a balloon which is deflating slowly, is decreasing at a rate of 2 cm per minute.

3

At what rate is the volume decreasing when the radius is 10 cm?
(Give your answer in terms of π)

(NOTE: The volume of a sphere is given by $V = \frac{4}{3}\pi r^3$)

(c) Find $\int \frac{1}{\sqrt{9-4x^2}} dx$

2

(d) Show that $\tan^{-1} \alpha + \tan^{-1} \beta = \tan^{-1} \left(\frac{\alpha + \beta}{1 - \alpha\beta} \right)$

3

For $0 < \alpha < 1$ and $0 < \beta < 1$

QUESTION 10: (10 Marks) Start a New Page

(a) Find the derivative of 5^x 1

(b) (i) Show that $1 + \frac{1}{2x-1} = \frac{2x}{2x-1}$ 1

(ii) Hence find $\int \frac{x}{2x-1} dx$ 2

(c) Find $\int \cot x dx$ 2

(d) (i) Find $\frac{d}{dx} \ln\{x + \sqrt{x^2 - 1}\}$ 2

(ii) Hence, or otherwise, find $\int_1^3 \frac{1}{\sqrt{x^2 - 1}} dx$, leaving your answer in exact form. 2

QUESTION 11: (10 Marks) Start a New Page

Marks

You are given the curve $y = \frac{x}{x^2+1}$

(i) Show that this is an odd function. 1

(ii) Show that the curve has turning points at $(1, \frac{1}{2})$ and $(-1, -\frac{1}{2})$ and describe their nature 3

You do not need to find any inflexion points.

(ii) Evaluate $\lim \left(\frac{x}{x^2+1} \right)$ 1

(iii) Sketch the curve, showing all the information you have just found above 2

(iv) Find the area under this curve and between the lines $x = 1$ and $x = 2$. 1

(v) With reference to your sketch, and your answer to part (iv) above, explain why 2

$$\ln \left(\frac{5}{2} \right) < 1$$

END OF THE EXAMINATION

SOLUTIONS

1/ B 2/ D 3/ B 4/ C 5/ A

Question 6:

(a) (i) $\frac{1}{3} \tan^{-1} \frac{x}{3} + k$ (ii) $-\frac{1}{3} \ln(1-3x) + k$

(iii) $\int \tan^2 x \, dx = \int (\sec^2 x - 1) \, dx$
 $= \tan x - x + k$

1 mark each
(no penalty for
no "k")

(b) (i) $\frac{x \cdot \frac{1}{x} - \ln x}{x^2} = \frac{1 - \ln x}{x^2}$

1 for either

(ii) $\cos x e^{\sin x}$

1 MARK

(c) $2 \sin^{-1} \frac{x}{2} \Big|_0^2 = 2 \sin^{-1} 1 - 2 \sin^{-1} 0$
 $= \pi$

1 for integral
1 MARK

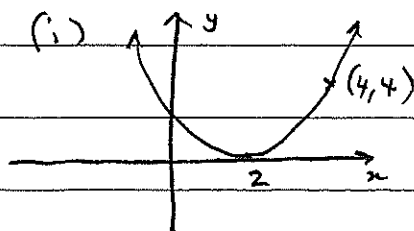
(d) $y' = 3x^2 e^{x^3}$
 $y'' = e^{x^3} \cdot 6x + 3x^2 \cdot 3x^2 e^{x^3}$
 $= 3x e^{x^3} [2 + 3x^3]$

1 MARK

2 MARKS

QUESTION 7:

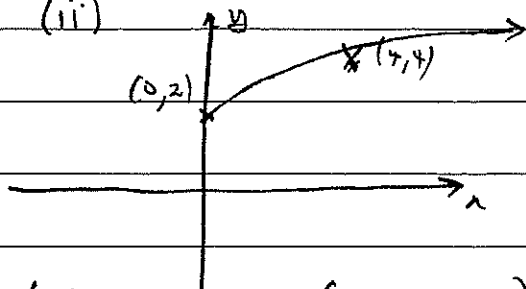
(a) (i)



$$Q: x \geq 2$$

1 mark

(ii)



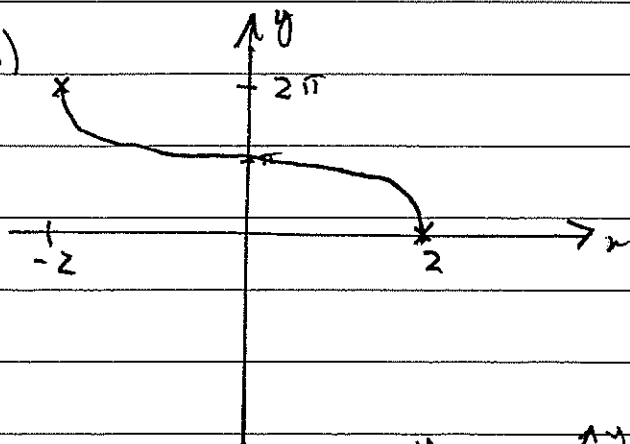
1 for shape
1 for (0, 2)

(iii)

$$Q: \{x: x \geq 0\} \quad R: \{y: y \geq 2\}$$

1 each = 2

(b)



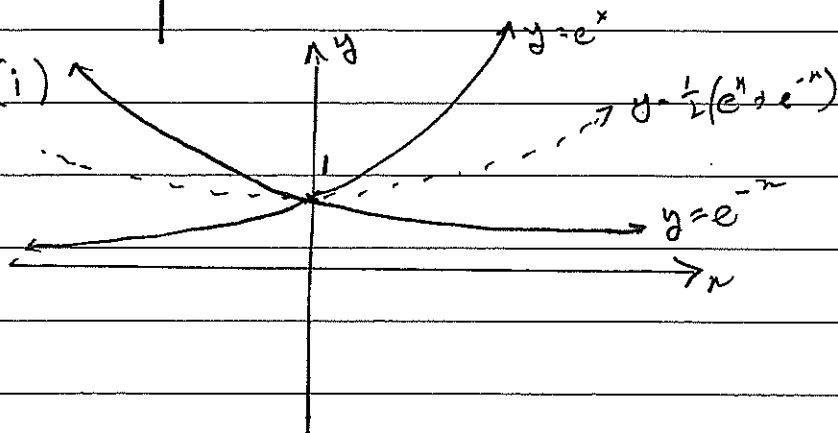
KEYPOINTS for
2 MARKS:

(i) (-2, 0) and (2, 0)

(ii) (0, 2\pi)

(iii) shape

(c) (i)



1 for each graph
= 3

QUESTION 8:

(a) $x+1 = e^5$

$$x = e^5 - 1$$

$$\approx 147.413$$

1 MARK

(must be to 3 dec places)

(b) $VOL = \pi \int_0^{2\sqrt{3}} \frac{1}{4+x^2} dx$

$$= \pi \frac{1}{2} \left[\tan^{-1} \frac{x}{2} \right]_0^{2\sqrt{3}}$$

$$= \frac{\pi}{2} \cdot \tan^{-1} \sqrt{3}$$

$$= \frac{\pi^2}{6}$$

1 for correct integral

2 MARKS

(c) (i) $\frac{dy}{dx} = \tan^{-1} x + x \cdot \frac{1}{1+x^2}$

1 MARK

(ii) $\int_0^1 \tan^{-1} x dx = \int_0^1 \left(\tan^{-1} x + \frac{x}{1+x^2} - \frac{x}{1+x^2} \right) dx$ ① for realising the connection

$$= x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) \Big|_0^1$$

① for integrals

$$= \tan^{-1}(1) - \frac{1}{2} \ln(2)$$

② for answer

$$= \frac{\pi}{4} - \frac{1}{2} \ln 2$$

(d) (i) $\frac{1}{\sqrt{1-x^2}} + \frac{-1}{\sqrt{1-x^2}} = 0$

1 MARK

(ii) This means that it is a constant.

The value is $\frac{\pi}{2}$

1 for "constant"

QUESTION 9:

(a) $\frac{dy}{dx} = \frac{d}{dx} \ln \sqrt{x-1} - \frac{d}{dx} \ln x$

$$= \frac{1}{2} \cdot \frac{1}{x-1} - \frac{1}{x}$$

$$= \frac{x - 2(x-1)}{2x(x-1)}$$

$$= \frac{2-x}{2x(x-1)}$$

← 1 for this

1 for simplification

(b) $\frac{dr}{dt} = 2$

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dr} = 4\pi r^2$$

$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$$

$$= 4\pi r^2 \cdot 2$$

← 1 MARK

← 1 MARK

At $t = 10$ $\frac{dV}{dt} = 800\pi$

1 MARK

(c) $\int \frac{1/3}{\sqrt{1 - \frac{4}{9}x^2}} dx$

$$= \frac{3}{2} \cdot \frac{1}{3} \sin^{-1} \frac{2x}{3} + k$$

$$= \frac{1}{2} \sin^{-1} \frac{2x}{3}$$

OR

$$\int \frac{1/2}{\sqrt{9/4 - x^2}} dx$$

$$= \frac{1}{2} \cdot \sin^{-1} \frac{x}{3/2}$$

$$= \frac{1}{2} \sin^{-1} \frac{2x}{3}$$

1 for arcosing $\int$

1 for answer

(d) Let $A = \tan^{-1} \alpha$ and $B = \tan^{-1} \beta$.

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$= \frac{\alpha + \beta}{1 - \alpha\beta}$$

$$\therefore A+B = \tan^{-1} \left(\frac{\alpha + \beta}{1 - \alpha\beta} \right)$$

3 MARKS.

QUESTION 10:

(a) $\frac{d}{dx}(5^x) = (\ln 5)5^x$

1 mark

(b) (i) $1 + \frac{1}{2x-1} = \frac{2x-1+1}{2x-1}$
 $= \frac{2x}{2x-1}$

1 mark

(ii) $\int \frac{2}{2x-1} dx = \frac{1}{2} \int \frac{2x}{2x-1} dx$
 $= \frac{1}{2} \int 1 + \frac{1}{2x-1} dx$

← 1 for this

$= \frac{x}{2} + \frac{1}{4} \ln(2x-1) + k$

1 mark (no penalty for no "k")

(c) $\int \cot x dx = \int \frac{\cos x}{\sin x} dx$

← 1 for realising

$= \log \sin x + k$

← 1 for answer

(d) (i) $\frac{1 + \frac{1}{2} \cdot 2x(x^2-1)^{-1/2}}{x + \sqrt{x^2-1}}$

← 1 mark

$= \frac{(x^2-1)^{-1/2} [(x^2-1)^{1/2} + x]}{x + \sqrt{x^2-1}}$

1 mark

$= (x^2-1)^{-1/2}$
 $= \frac{1}{\sqrt{1-x^2}}$

(ii) $\int_1^3 \frac{1}{\sqrt{x^2-1}} dx = \ln \{x + \sqrt{x^2-1}\} \Big|_1^3$

← 1 for realising

$= \ln(3 + \sqrt{8}) - \ln(1 + 0)$

$= \ln(3 + \sqrt{8})$

$= \ln(3 + 2\sqrt{2})$

1 for either answer

QUESTION 11:

(i) $f(a) = \frac{a}{a^2+1}$

1 MARK

$f(-a) = \frac{-a}{(-a)^2+1}$

$= -f(a) \therefore \text{ODD}$

(ii) $\frac{dy}{dx} = \frac{(x^2+1) - x \cdot 2x}{(x^2+1)^2}$

$= \frac{1-x^2}{(x^2+1)^2}$

At T.P.'s $\frac{dy}{dx} = 0$

$\therefore \begin{cases} x=1 & \text{or } x=-1 \\ y=1/2 & y=-1/2 \end{cases}$

1 for each point
= 2

$\begin{array}{c|c|c|c|} x & 0 & 1 & 2 \\ \hline y' & + & 0 & - \end{array}$

$\therefore \text{max at } (1, 1/2)$

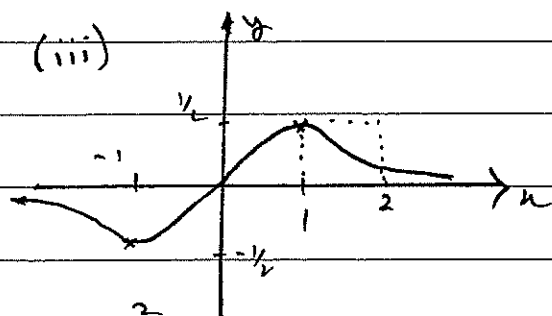
$\begin{array}{c|c|c|c|} x & -2 & -1 & 0 \\ \hline y' & - & 0 & + \end{array}$

min at $(-1, -1/2)$

1 for identification

(iii) $\lim_{x \rightarrow \infty} \left(\frac{x}{x^2+1} \right) = 0$

1 MARK



2 MARKS

KEYPOINTS: both T.P.'s
both asymptotes.

(iv) $\int_2^5 \frac{x}{x^2+1} dx = \frac{1}{2} \ln(x^2+1) \Big|_2^5$

$= \left(\frac{1}{2} \ln(5) - \frac{1}{2} \ln(2) \right)$
 $= \frac{1}{2} \ln(5/2)$

} 1 for either

(v) The area of the dotted rectangle above is $\frac{1}{2} \text{ m}^2$ which is greater than the actual area

ie $\frac{1}{2} > \frac{1}{2} \ln(5/2)$

} 1 MARK

1 MARK

$\therefore \ln(5/2) < 1$